



Catch Horizons on BBC

Horizons Series 5, part 2 is scheduled on BBC World News from October 3rd every Saturdays at 7 am, 2 pm and Sundays at 8 pm.



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Imagining the future

In an exclusive guest post for Digit, the presenter of BBC's award winning show Horizons, Adam Shaw, tells us about discovering the latest technologies and innovations that could revolutionise the way we live

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Imagining the future is what Horizons is all about. This series we have looked at some of the biggest challenges facing our planet in the 21st century, from energy generation to food production and beyond. We explore how scientists across the world are rising to this challenge and developing the technology of the future.

The technology of the future doesn't necessarily have to be invented or born out of thin air. Janine Benyus, the founder of the Biomimicry Institute, explained to us this series that nature is the greatest laboratory of all – understanding and learning from it – she says, is already bringing us huge advances. If you think about the world as a scientific core which, through evolution and various other processes, has been trying to solve scientific problems for thousands of years, it's done the work for us.

Biomimicry

Biomimicry is perhaps one of the most appealing areas of science; not just because it makes for exciting television, but because it is something that everyone can understand and has a connection to. Quantum mechanics, astrophysics, nuclear biology etc., are all really exciting areas of science but they're very distant from ordinary people's everyday experiences. Biomimicry, however, takes its

inspiration from nature, so you can look all around the world and see inspiration.

The most obvious example of biomimicry is Velcro. When the Swiss engineer, George de Mestral, came back from walking his dog on the moors he noticed it was covered in these burrs which he couldn't get off. When de Mestral examined the seedpods under a microscope he marvelled at how they bristled with hooks



Alex Forrest and Prof Gordon Cheng: enabling Biomimicry.

ingeniously shaped to grasp at animal fur. The plant had worked out a system of dispersing its seeds by just attaching themselves to any passing animal. He realised that being able to attach something really firmly and then detach it would be really useful and with that, he invented Velcro and changed the world of fastening for ever.

Triboelectricity

The future of energy generation and supply is one of those fundamental forces that will shape our societies. For the most part we think of energy generation as big energy plants, solar fields, nuclear power stations or gas and coal power stations, but energy is all around us. We use energy

as we walk and generate energy all day, even when our clothes rub together they generate energy, we just don't have a way of gathering that.

In this series of Horizons we speak to Professor Zhong Lin Wang, from the Chinese Academy of Science in Beijing, who is researching a fascinating new type of electricity generation called the triboelectric effect. The word comes from the Greek word 'tribein' meaning 'to rub'.

Triboelectricity can create a surprising amount of power by rubbing or touching two different materials together and Professor Wang is looking at ways to collect this energy and use it. For instance on the floor of a shopping centre; when your foot touches the floor, the movement can generate a tiny electrical current. If you can capture that tiny current in a space where hundreds or thousands of people walk every day, suddenly you're generating quite a lot of electricity. The same is true with clothing; scientists are experimenting with putting collectors on clothing and as you walk, your clothes rubbing together will create an electric charge that you could attach to a mobile phone in your pocket and have it constantly charging.

This use of movement to generate electricity has been seen before; a couple of years ago a hotel put their gym on the grid for a publicity stunt. The idea was that if you went to the gym and used the bicycle you generated electricity for the hotel and they knocked a bit of money off your bill. This research is taking that idea further and, rather than seeing us as consumers of electricity, each person becomes a generator.

I think triboelectricity is one of those things which is really emblematic of Horizons, this isn't something that's going to happen tomorrow but it is fundamental